

LOTUS RETAIL ANALYTICS

Business Intelligence & Data Analytics Project

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EXECUTIVE SUMMARY

TheLotus RetailAnalytics project transforms raw transactional data into actionable business

intelligence through Power BI visualization and advanced data analytics. This report documents the complete journey from raw data preparation through strategic business recommendations.

Key Project Metrics	
Metric	Value
Total Orders Analyzed	12,000 transactions
Analysis Period	2022-2024 (3 years)
Average Order Value	3,779 EGP per order
Data Tables	4 Dimension + 2 Fact Tables
Key KPIs Tracked	Revenue, Cost, Profit, Margin, Orders

Project Objectives

- ☐ Transform raw operational data into strategic business intelligence
- ☐ Enable data-driven decision-making across retail operations
- ☐ Provide multi-dimensional analysis of sales performance
- ☐ Identify high-performing and underperforming business units
- ☐ Generate evidence-based recommendations for operational improvement

1. PROJECT OVERVIEW

The project is a retail sales analysis using Power BI, designed to transform raw operational data

into business intelligence and actionable insights. The fundamental approach is to take data from multiple operational tables, prepare it using Power Query, organize it through a data model, and visualize it in an interactive dashboard.

1.1 Project Scope

- ☐ Data Source: Transactional retail data from 2022-2024
- ☐ Data Volume: 12,000 orders across multiple stores and regions
- ☐ Data Preparation: Power Query cleaning and standardization
- ☐ Data Modeling: Star-schema dimensional model
- ☐ Analytics: Business calculations and KPI measures
- ☐ Visualization: Interactive dashboard with multiple perspectives
- ☐

2. DATASET STRUCTURE

The original data is divided into Dimension Tables (reference data) and Fact Tables

(transaction data), enabling multi-dimensional analysis.

2.1 Dimension Tables

Table Name	Purpose
dim_customers	Customer data and identifiers
dim_products	Product names, prices, and categories
dim_stores	Store locations and regional assignments
dim_employees	Employee information

2.2 Fact Tables

fact_sales	Purpose
fact_orders	Revenue, cost, and profit transactions
	Order details and quantities

3. DATA CLEANING & POWER QUERY

Before analysis, data was prepared within Power Query. The Applied Steps in the project are:

Power Query Step	What We Did
Source	Loaded the raw data into Power BI
Navigation	Identified and selected necessary tables
Promoted Headers	Converted first row to column names
Changed Type	Modified data types (Text, Numbers, Dates)
Lowercased Text	Standardized text for consistency
Removed Duplicates	Eliminated duplicate records
Filtered Rows	Removed incomplete or invalid data

3.1 Data Quality Improvements

- Consistency: Standardized text and formatting
- Completeness: Removed error rows
- Integrity: Assigned correct data types
- Accuracy: Filtered out test and incomplete data

4. DATA MODEL ARCHITECTURE

After data cleaning, the Fact Tables and Dimension Tables were organized in a data model that allows linking sales data to product, store, and customer information.

4.1 Star Schema Structure

- ❑ Central Fact Tables: Contain transactional data
- ❑ Surrounding Dimension Tables: Provide context
- ❑ Relationships: Connect facts to dimensions

5. BUSINESS CALCULATIONS & KPIs

After data cleaning and modeling, Power BI calculations convert raw fields into meaningful business metrics.

5.1 Core KPIs

Total Revenue	Business Meaning
Total Cost	Total income from all sales
Total Profit	Total cost of goods sold
Profit Margin %	Profit = Revenue - Cost
Total Orders	Percentage of revenue that becomes profit
Total Customers	Count of transactions/orders
Average Order Value	Count of unique customers
	Average revenue per order (~3,779 EGP)

6. DASHBOARD STRUCTURE

The Dashboard converts calculations into visual business information. The interactive Power BI dashboard enables users to explore data from multiple perspectives using slicers and filters.

6.1 Dashboard Components

- ❑ KPI Cards: Display Revenue, Profit, Orders, AOV, Customers
- ❑ Time Analysis: Revenue trends over months and quarters
- ❑ Store Performance: Store-level revenue and profit comparison
- ❑ Category Analysis: Product category contribution
- ❑ Regional Analysis: Geographic performance
- ❑ Slicers/Filters: Filter by Year, Region, Category

7. ANALYSIS & BUSINESS INSIGHTS

The dashboard is designed to use numbers to understand performance. Key types of insights include:

7.1 Performance Analysis

- ☐ Time-Based: Revenue trends across months and quarters
- ☐ Category: Products contributing largest share of revenue
- ☐ Store Level: Revenue and profit differences by location
- ☐ Regional: Revenue comparison between regions
- ☐ Customer: Average order value and order frequency

8. STORE PERFORMANCE ANALYSIS

Store-level analysis reveals performance variation across locations and identifies improvement opportunities.

8.1 Key Findings

- ☐ Revenue distribution varies significantly across stores
- ☐ High-revenue stores may not always have highest margins
- ☐ Profitability depends on both volume and cost management
- ☐ Performance patterns indicate operational best practices

9. REGIONAL PERFORMANCE ANALYSIS

Geographic analysis shows how different regions contribute to overall business performance.

9.1 Key Findings

- ☐ Revenue contribution varies by region
- ☐ Regional profitability may differ from volume
- ☐ Market conditions or operational differences drive results
- ☐ Regional performance guides market development strategy

10. PRODUCT CATEGORY ANALYSIS

Category analysis identifies which products contribute most to revenue and profit.

10.1 Key Findings

- ☐ Revenue is concentrated in key categories
- ☐ High-revenue categories deserve priority in inventory
- ☐ High revenue does not guarantee high profit
- ☐ Category trends inform product strategy

11. BUSINESS RECOMMENDATIONS & ACTIONS

Each recommendation follows: Insight → Business Implication → Recommended Action

11.1 Key Recommendations

- High-Revenue Categories: Prioritize inventory and marketing investment
- High-Revenue, Low-Profit Stores: Review costs, pricing, and product mix
- Underperforming Regions: Analyze demand and marketing effectiveness
- Low AOV Opportunities: Implement upselling and cross-selling
- Seasonal Planning: Use trends to improve inventory planning

12. CONCLUSION & FUTURE IMPROVEMENTS

12.1 Project Achievement

This project transforms raw transactional data into a Business Intelligence dashboard. The fundamental value is moving from having data to having clear, actionable information.

12.2 Key Deliverables

- ☐ Cleaned and prepared dataset through Power Query
- ☐ Organized data model connecting tables
- ☐ Comprehensive set of business KPI measures
- ☐ Interactive Power BI dashboard
- ☐ Evidence-based insights and recommendations
- ☐ Professional documentation

12.3 Future Enhancements

- ☐ Advanced forecasting models
- ☐ Drill-through capabilities
- ☐ Customer analytics and segmentation
- ☐ Return and quality analysis
- ☐ Automated alert systems
- ☐ Marketing attribution measurement